



REPORT

Assignment 1: SIR model (ODE)

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1 Introduction

Epidemic preparedness is a key challenge of our time, as outbreaks can impact our highly interconnected world on health, economic, and social levels, as demonstrated by the COVID-19 pandemic. As stated by the Global Preparedness Monitoring Board, organized by the World Health Organization (WHO) and the World Bank (WBG), preventing and preparing for the next pandemic requires adapting to changing risks, protecting communities and strengthening cross-sector connections ([2]). These principles are grounded in the variety of models and frameworks that describe the evolution of epidemics and pandemics over time. Such models enable researchers and policy makers to make prediction and plan responses to infectious diseases outbreaks.

Modeling these diseases is particularly challenging given the complexity of the phenomenon and the many factors influencing transmission, such as demographics, social behavior, contact structure, and environmental conditions. While models can always be expanded to incorporate more realism, even the simplest epidemiological frameworks, such as the basic SIR model, offer valuable insight, such as determining whether an infection leads to an epidemic, how fast it spreads, and how long it persists ([5]).

This report explores how to build on the simplest SIR model and to extend the analysis to more complex scenarios, while identifying appropriate parameter values and simulation settings. In particular, it is structured as follows: section 2 describes the theoretical background of the models used, section 3 outlines the numerical methods implemented for solving the systems, section 4 presents the simulations and results, and section 5 concludes.

2 Theory

The SIR model, first described in Kermack and McKendrick ([5]), is a deterministic, continuous-time model represented by a set of ordinary differential equations (ODEs). It captures the dynamics of acute infections, where the pathogen causes illness for a limited period of time followed by (typically lifelong) immunity. The model, in its basic form, divides the population into three compartments:

- Susceptible (S): individuals who are healthy but can be infected;
- Infected (I): individuals currently infectious;
- Recovered (R): individuals who have recovered and gained immunity.

Movements between compartments occur at constant or time-varying rates and can capture increasingly complex scenarios, such as changes in population size and subgroups, seasonality, or preventive strategies. This makes SIR-based models highly adaptable to multiple diseases and epidemiological settings.

2.1 The Basic SIR model

In the basic SIR model, the population is closed (population size N is constant) and divided into the three groups mentioned above. Therefore, the proportions of susceptible, infected and recovered individuals over N always sum up to one (i.e. $S + I + R = 1$).

Infected individuals take time $1/\gamma$ to recover, so that the transition from infected to recovered happens at constant per-capita rate γ . From S to I , the transition is governed by the force of infection λ , which reflects the probability of acquiring the infection through contact:

$$\lambda = \beta I,$$

where β is the transmission rate, defined as the product of the contact rate and the probability of transmission per contact. For the whole population, the rate of new infections is βIS .

The system of ODEs describing the dynamics is:

$$\begin{aligned}\frac{dS}{dt} &= -\beta SI, \\ \frac{dI}{dt} &= \beta SI - \gamma I, \\ \frac{dR}{dt} &= \gamma I.\end{aligned}$$

2.1.1 Threshold Phenomenon

Starting from these basic SIR equations, [5] identified a particular threshold depending on the proportion of susceptible individuals and the infectivity, recovery (and death) rates that determines whether an infection spreads. This relation constitutes the so-called basic reproductive ratio, which can be interpreted as the average number of secondary cases arising from an average primary case in an entirely susceptible population ([4]), and it is defined as

$$R_0 = \frac{\beta}{\gamma}.$$

We can show that for an epidemic to decline, $dI/dt = I(\beta S - \gamma) < 0$ implies $S < \frac{\gamma}{\beta} = \frac{1}{R_0}$. In a fully susceptible population ($S(0) = 1$), the epidemic dies out if $S = 1 < \frac{1}{R_0}$, hence $R_0 < 1$. If $R_0 > 1$, the number of infected individuals initially grows, leading to an epidemic. This threshold condition also holds for more complex extensions of the basic SIR model.

2.2 The SIR Model with Demography

The basic SIR model captures well the characteristics of infections that are expected to spread rapidly and end over time. However, over longer time horizons, births and natural deaths intervene in the population dynamics. In the SIR model with demography (and constant N), new births enlarge the susceptible pool at rate μ , while individuals have a life span of $1/\mu$ and die at the same per-capita rate in all compartments (ensuring rate of inflow = rate of outflow, so constant population). The model equations become:

$$\begin{aligned}\frac{dS}{dt} &= \mu - \beta SI - \mu S, \\ \frac{dI}{dt} &= \beta SI - \gamma I - \mu I, \\ \frac{dR}{dt} &= \gamma I - \mu R.\end{aligned}$$

Accounting for mortality in the infectious compartment, the reproductive ratio becomes: $R_0 = \frac{\beta}{\gamma + \mu}$. And as seen above, if $R_0 < 1$, then the disease-free equilibrium is reached. However, this system also admits an endemic equilibrium, where the disease persists at constant prevalence, if $R_0 > 1$, given by:

$$(S^*, I^*, R^*) = \left(\frac{1}{R_0}, \frac{\mu}{\beta}(R_0 - 1), 1 - \frac{1}{R_0} - \frac{\mu}{\beta}(R_0 - 1) \right).$$

2.3 The SIR Model with Mortality

Some diseases carry a mortality risk during infection. This can be modeled by introducing a probability ρ of dying while infected. The system of ODEs becomes:

$$\begin{aligned}\frac{dS}{dt} &= \nu - \beta SI - \mu S, \\ \frac{dI}{dt} &= \beta SI - \frac{\gamma + \mu}{1 - \rho} I, \\ \frac{dR}{dt} &= \gamma I - \mu R.\end{aligned}$$

Note that in this case, the total population size is no longer constant, and the endemic equilibrium levels are shifted, however it is generally feasible and stable as long as $R_0 > 1$ ([4]). Moreover, the parameter ρ accelerates the decline of I , and could lead to complete removal of the host population if mortality is very high.

2.4 The SEIR Model

Some diseases are best modeled by accounting for a latent period, when the pathogen reproduces rapidly within the host but is relatively unchallenged by its immune system. In this case, it is useful to introduce the exposed category (E), which means individuals who are infected but not yet infectious. The SEIR model equations are:

$$\begin{aligned}\frac{dS}{dt} &= \mu - \beta SI - \mu S, \\ \frac{dE}{dt} &= \beta SI - (\mu + \sigma)E, \\ \frac{dI}{dt} &= \sigma E - (\gamma + \mu)I, \\ \frac{dR}{dt} &= \gamma I - \mu R,\end{aligned}$$

where $1/\sigma$ is the mean latent period, hence σ is the rate from exposed to infected. The SEIR model slows the initial growth of infections, since individuals must first pass through the exposed stage before contributing to transmission.

2.5 Seasonal Forcings

Finally, we account for diseases that exhibit seasonal behavior due to environmental or population behavioral factors. Seasonality is modeled by allowing the transmission rate to vary with time. β is then re-defined as a function of time:

$$\beta(t) = \beta_0(1 + \beta_1 \cos(\omega t)),$$

where β_0 is the baseline transmission rate, β_1 is the amplitude (strength) of seasonal forcing, and ω corresponds to the angular frequency, i.e. how often cycles repeat (e.g., yearly, monthly, weekly).

By varying the amplitude β_1 (and ω), we explore how stronger or weaker seasonal forcing affects the persistence and recurrence of outbreaks.

2.6 Vaccination and Preventive Strategies

Epidemic preparedness requires studying and implementing preventive strategies, such as vaccination, which reduces the number of susceptible individuals in the population, and quarantining and isolation, which reduce the number of infected individuals freely mixing in the population. These interventions are modeled by modifying the flow between population compartments, to compare them with unmodified settings.

3 Numerical methods

3.1 ODE Solvers

The non-linear system of ODEs representing the SIR model cannot generally be solved explicitly. While [3] provided a parametric analytical solution for the SIR model with equal birth and death rates, more commonly approximate solutions are found through numerical integration methods.

The simplest approach to numerical integration is the Euler method, which translates differential equations into discrete-time form by updating the solution at time $t + \Delta t$ using the rate of change at time t , multiplied by the step size Δt :

$$X(t + \Delta t) = X(t) + \Delta t \cdot \frac{dX}{dt}(t).$$

However, this method leads to a global error accumulation that scales as $\mathcal{O}(\Delta t)$. This means large errors accumulate linearly with the step size, requiring very small time steps to maintain accuracy.

In this report, to improve accuracy, we used the `solve_ivp` solver from the `scipy.integrate` library in Python, which implements by default a Runge–Kutta method of order 5(4) (denoted as RK45). Improving upon Euler’s method, Runge–Kutta methods estimate the rate of change at several intermediate points and combine them into a weighted average. RK45 in particular uses both 4th- and 5th-order solutions computed at each step, with the difference between them providing an error estimate so that the final global error scales with $\mathcal{O}(\Delta t^5)$, improving our solution accuracy.

The `solve_ivp` solver requires: the system of ODEs (in our case, the SIR model or its extensions), initial conditions S_0 , I_0 and R_0 , so that they sum up to 1 (usually, $S_0 = 0.99$, $I_0 = 0.01$ and $R_0 = 0$, corresponding to a mostly susceptible population with a small initial outbreak), and, finally, a time grid (in days, or years). Solutions were obtained over both short-term horizons (e.g. 14 days in the influenza outbreak case) and long-term horizons (up to decades, to analyze demographic and seasonal effects).

The numerical solutions were then visualized both as time series plots, for $S(t)$, $I(t)$ and $R(t)$; and as phase space diagrams (S vs. I) using the `matplotlib.streamplot` command to show the trajectories of the dynamics.

3.2 Parameter Estimations

In addition to solving the ODEs numerically, part of the analysis required estimating parameters (β, γ) from observed outbreak data. This was done by minimizing the difference between the model’s predicted number of infections and the historical data using a least squares approach.

Formally, given observed data $I_{\text{obs}}(t)$ and the model-predicted infections $I_{\text{model}}(t, \beta, \gamma)$, we minimized the residual sum of squares:

$$\text{RSS}(\beta, \gamma) = \sum_t (I_{\text{obs}}(t) - I_{\text{model}}(t, \beta, \gamma))^2.$$

This fitting procedure was implemented via the command `scipy.optimize.least_squares`.

3.3 Fourier Analysis

When demographic changes (births and deaths) are included, the model can exhibit damped oscillations around an endemic equilibrium. To study this oscillatory behavior, we apply a discrete Fourier transform to the infected trajectory $I(t)$, using the `numpy.fft` library. The command implements a Fast Fourier Transform (FFT), an algorithm that converts the time-domain signal into the frequency domain, producing a spectrum that identifies which frequencies dominate the oscillations. The magnitude spectrum provides information on the recurrence of epidemics, damping effects, and on the stability of the endemic equilibrium.

4 Results and discussion

4.1 Numerical Integration of the SIR Model

4.1.1 Phase Spaces and the role of R_0

We begin with the basic specification of the SIR model in a closed population, where the spread of infection goes in one direction from susceptible to infected at rate β , and from infected to recovered, at rate γ . Once recovered, an individual can no longer become susceptible, gaining lifelong immunity.

We consider one scenario where the epidemic spreads, and one where it does not. As stated above, this can be done by varying the threshold ratio $R_0 = \frac{\beta}{\gamma}$. We first set the average infection period $\frac{1}{\gamma} = 14$, an appropriate value for acute infections, and then we consider β according to the scenario we want to plot. We also set initial condition $S(0) = 0.99, I(0) = 0.01, R(0) = 0.0$ to model an infection at the beginning of the spread, and obtain these two plots overtime:

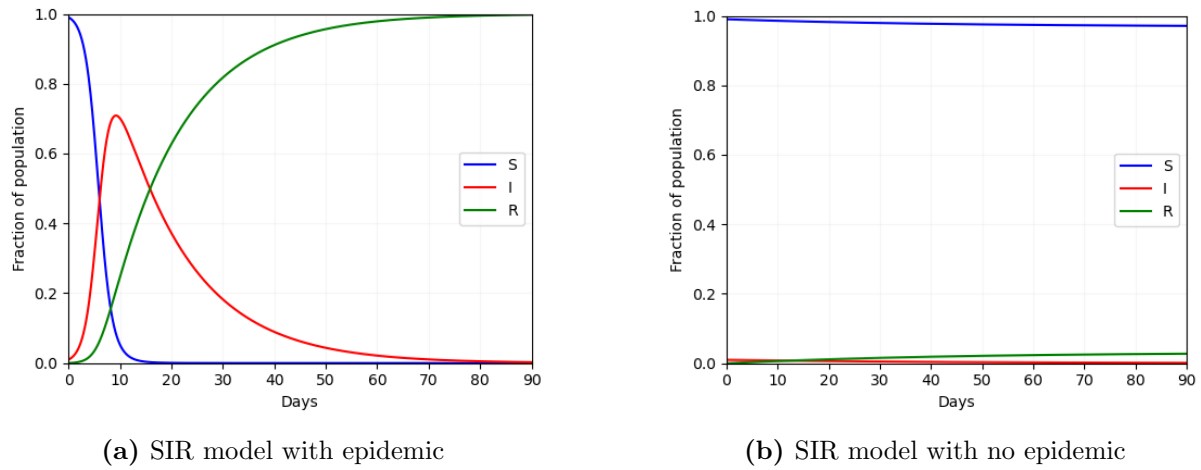


Figure 1: SIR dynamics evolution overtime, starting from $S(0) = 0.99, I(0) = 0.01, R(0) = 0.0$. In the first plot, $\beta = 0.85$ and $\frac{1}{\gamma} = 14$, so that $R_0 = \frac{\beta}{\gamma} = 11.90$, while in the second one, $\beta = 0.01, \frac{1}{\gamma} = 14$, so that $R_0 = 0.7$, so that the disease does not spread.

For the phase space diagrams, representing the epidemic trajectories for multiple initial conditions, we will study the system in the (S, I) plane. This because $S + I + R = 1$, and $R = 1 - S - I$ can be determined.

As shown in the first graph, in the epidemic case, the infected proportion initially increases, peaks, and eventually decays to the disease-free state, i.e. $(S^*, I^*, R^*) = (0, 0, 1)$, where all individuals are recovered and immunized. If the threshold condition is not satisfied, the number of infected individuals declines immediately, as shown by the trajectories collapsing vertically toward the x-axis.

This behavior is consistent with the time series plots (Figure 1), which show that when $R_0 < 1$, the epidemic never takes off.

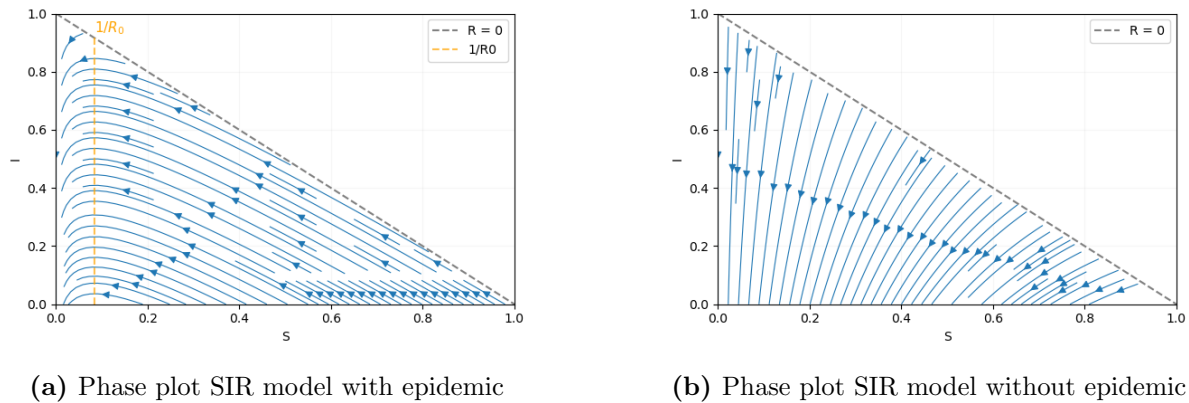


Figure 2: Arrowed lines represent trajectories of the epidemic dynamics for different initial conditions. The threshold condition is represented by the vertical dotted line in the figure where $S(0) > \frac{\gamma}{\beta} = \frac{1}{R_0}$. The dashed line set the feasibility boundary $S + I = 1$.

4.1.2 Estimating Parameters

Parameters β and γ can be estimated by fitting the SIR model to observed data. As an example, we consider a real world case of an influenza outbreak in a boy's school with a total of 763 students. Assuming that one infected boy started the epidemic on day 0, we estimate the parameters fitting the SIR model to the data using a least squares procedure. Minimizing the residuals between the empirical infection curve and model predictions we obtain as parameters $\beta = 1.66$ and $\gamma = 0.45$. Then, $R_0 = 3.72 > 1$, confirming that the dynamics correspond to an epidemic scenarios. The fitted curve (Figure 3a) matches the rise and fall of infections, supporting the validity of the SIR framework in this setting (i.e., an acute infection in a closed population).

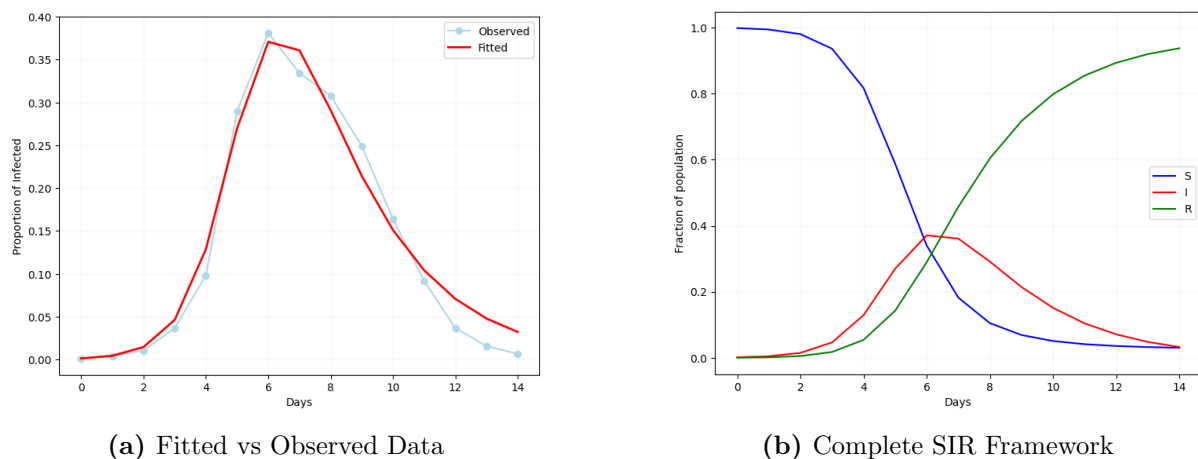


Figure 3: The first plot represents the fitted curve, given the estimated parameters $\beta = 1.66$ and $\gamma = 0.45$, while the second one contextualizes the infected curve into a complete SIR framework.

4.1.3 Vaccination Strategies

The influenza case at school provides a good (closed and limited) setting to explore preventive strategies, in particular vaccination, and evaluate its efficacy for outbreak control.

First, we consider a series of cases where vaccination is administered before the school term begins. In general, vaccination lowers the susceptible individuals pool, since vaccinated people move to the recovered group, once immunity is developed. Influenza vaccine could be set as a

mandatory requirement for students to attend lessons. Ideally, if 100% of coverage was reached, influenza would not spread. However, influenza vaccine is known to have a certain efficacy threshold, between 40 and 60% according to [7]. In our model, we set a vaccine efficacy of 55% (around the average value), so that more than half of the students develop immunity before the school term (and are "recovered"). As shown by Figure 5a, infection dynamics are not completely canceled out, but are strongly reduced.

In a more realistic but still optimistic setting, where vaccination is not mandatory, but potentially highly recommended, 60% of students get vaccinated before influenza season. This could be reached through convincing communication and awareness campaigns (although [1] underscores that the average vaccination rate for young people in the European country is at most around 30 %). In this case, the reduction in susceptibility is of 33.3% and the outbreaks dynamics are slowed down. Compared to the initial setting, where infectious peak was reached on day 6 with 282 infected students, if a third of the students vaccinates, then infections peak on day 10, when 119 individuals are affected.

Alternatively, we consider a scenario where vaccination is introduced as a reaction to the detection of the first case of influenza. For example, starting from day 1, after the first boy is sick, students get vaccinated at a constant rate of 20 students per day, moving to the "recovered" section. Then, the outbreak still slows down, with 206 individuals infected on day 7. Also, from plot 4a, we can see that the susceptible individuals pool reaches 0 when all remaining students are vaccinated (on day 11), while the recovered/vaccinated curve exhibits a rapid growth (plot 4c).

However, in real life the effectiveness of this reactive vaccination approach may be limited by two key factors: the delay in identifying the first influenza case and the time required for immunity to develop post-vaccination. As a result, its impact on outbreak mitigation is less pronounced than the first strategy.

In general, simulations demonstrate that higher vaccination coverage substantially lowers the epidemic peak and reduces the total number infected, potentially blocking epidemic spread. And this can be reached also without vaccinating 100% of individuals as plot 5a shows, obtaining the so-called herd immunity.

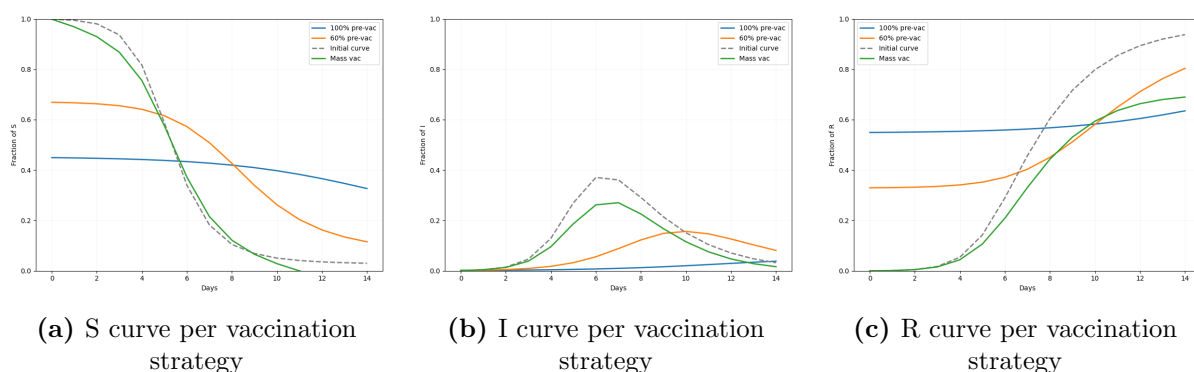


Figure 4: Each graph illustrates the S, I, and R curves for each vaccination strategy, compared to the baseline model (dashed lines). Overall, the curves demonstrate slower rates of increase across all scenarios, with a particularly notable reduction in the number of infected students.

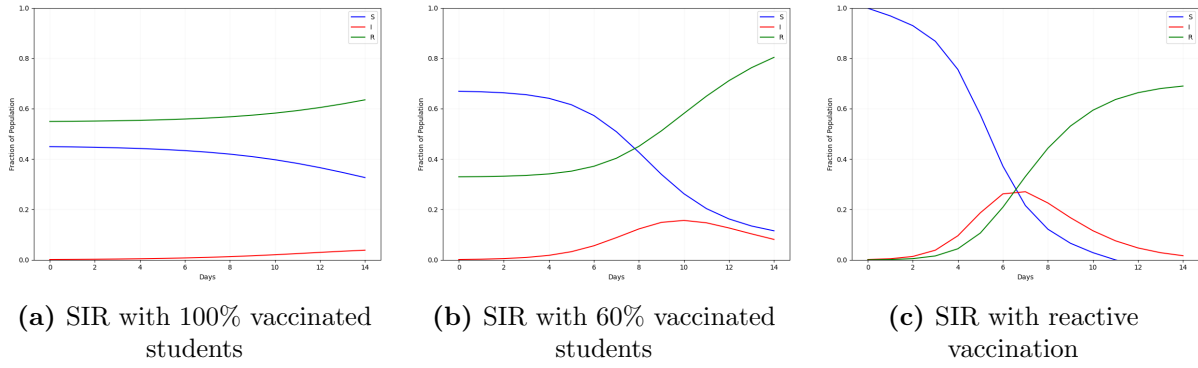


Figure 5: The complete SIR dynamics for the three vaccination strategies reveal that in the first scenario (Fig. 5a), the number of recovered individuals initially surpasses susceptible ones, effectively preventing disease spread, while in the other scenarios, even though transmission still occurs, the dynamics are significantly slowed.

4.2 Demography

4.2.1 Oscillatory Behavior to Equilibrium

We extend the basic SIR model to show the existence of another equilibrium, the endemic state, beyond the disease-free one. We introduce a birth-and-natural-death rate μ set at $(1/(70 \cdot 365))$ (since modeling daily dynamics). This implies an average lifetime of $1/\mu = 70$ years. While the susceptible pool is replenished at a rate μ , individuals die in all compartments at the same rate, so that population size N is kept constant. γ and β are set by considering the average infection period and the desired R_0 respectively, as above.

Phase space diagrams clearly highlight how the dynamics change compared to the basic SIR model. If $R_0 = \frac{\beta}{\gamma + \mu} > 1$, (where $\frac{1}{\gamma + \mu}$ is the average time now spent in the infected class), then the system converges to an endemic equilibrium $(S^*, I^*) = (\frac{1}{R_0}, \frac{\mu}{\beta}(R_0 - 1))$. On the other hand, if $R_0 \leq 1$, trajectories lead to a disease-free equilibrium of $(S^*, I^*) = (1, 0)$, sustained by the birth rate feeding into the susceptible pool.

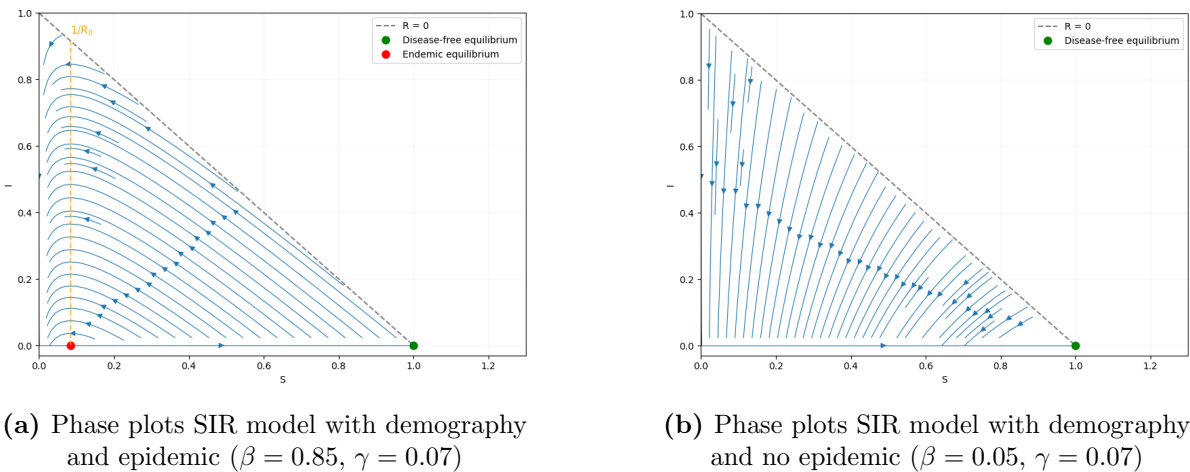


Figure 6: Arrowed lines represent trajectories of the epidemic dynamics for different initial conditions. Dashed line marks the feasibility boundary $S + I = 1$, while the dotted vertical line shows the epidemic threshold $S(0) > \frac{\gamma + \mu}{\beta} = \frac{1}{R_0}$. The points highlight the feasible equilibria.

We focus on the stability around the endemic equilibrium. Starting from an initial condition close to the equilibrium I^* , the infected fraction $I(t)$ exhibits damped oscillatory behavior around this equilibrium in the long run. From linear stability analysis ([6]), this occurs when

the Jacobian of the linearized system has negative trace, positive determinant, and complex eigenvalues. From the eigenvalue formulas, we can derive both the oscillation frequency and period, which we compare against estimates obtained by Fourier analysis of the $I(t)$ trajectory.

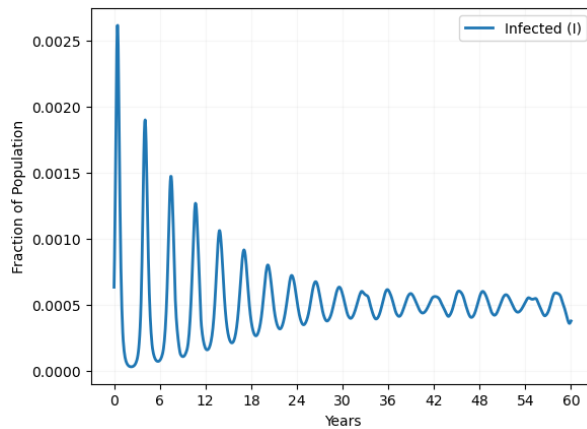


Figure 7: Starting from $I^* + \text{numpy.random.rand()} * 1e - 4$, $I(t)$ oscillates around the equilibrium I^* overtime. The oscillations are damped, meaning that the endemic equilibrium represented in fig. 6a in is a stable one, according to [6].

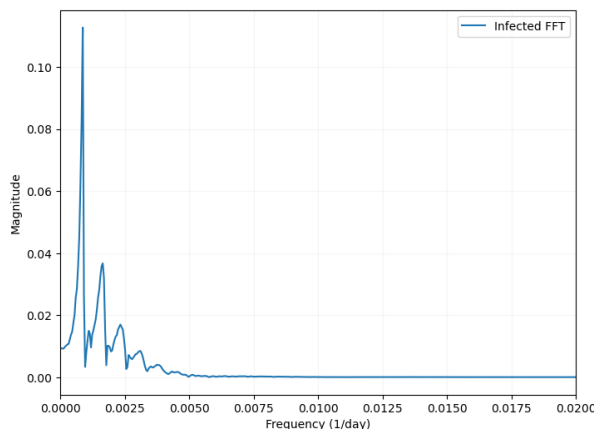


Figure 8: After applying the Fourier transform, we obtain the most frequent oscillation, i.e. Most prominent infection frequency: 0.00087 (1/day), for a Period ≈ 1153.79 days (3.16 years)

4.2.2 Infection-induced Mortality

Mortality can also result directly from the infection itself. To include this infection-induced mortality in the model, we add a probability ρ of dying while infected. This modifies the outflow from the infected category: instead of all infected individuals eventually recovering, a fraction ρ die and are removed from the system. As a result, the total population size N is not constant anymore.

The model is simulated by numerically integrating the ODEs over the absolute numbers of individuals, , as suggested by [4], starting from an almost completely susceptible population ($S(0) = 9900, I(0) = 100, R(0) = 0$). For visualization, results are normalized and presented in proportions.

Figure 9 shows that as the probability ρ increases, more infected individuals die rather than recover, reducing the peak of infection. This is also reflected in the dashed line for the

total population size N , which decreases faster with higher ρ . In the extreme case $\rho \rightarrow 1$, the epidemic rapidly collapses, since transitions between compartments are cut short by infection induced deaths.

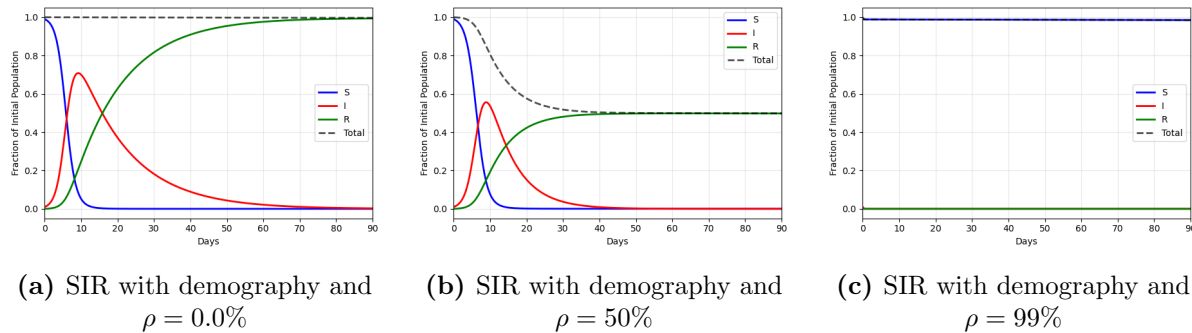


Figure 9: Epidemic dynamics with infection-induced mortality. As ρ increases, more infected individuals die instead of recovering, and the total population N (dashed line) declines faster. For $\rho \rightarrow 1$, the epidemic collapses quickly.

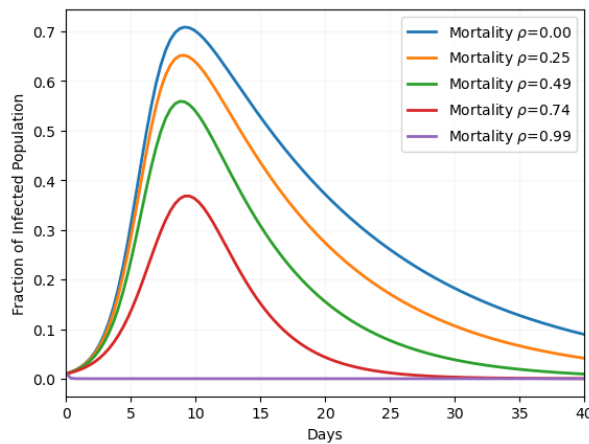


Figure 10: Effect of infection-induced mortality on the infected population. As ρ increases, the decline of I accelerates and the epidemic collapses earlier.

4.3 Variants of the SIR model

4.3.1 The SEIR Model

Another way in which SIR dynamics are changed is when modeling diseases that involve a latent period, during which individuals are infected, but not yet infectious. This is done by adding an Exposed (E) compartment, leading to the SEIR model. In this model, susceptibles become exposed after contact with an infectious person at rate proportional to β , while exposed become infected after $\frac{1}{\sigma}$ time (average latent period), i.e. at rate σ . Finally, infected individuals move into the recovered class at rate γ .

In our simulations, we fix the infectious period to 7 days (i.e., $\gamma = 1/7$) and vary the latent period by adjusting σ . Introducing the exposed class slows down the epidemic compared to the basic SIR model, since newly infected individuals must first pass through E before contributing to transmission. As σ decreases (longer latent period), the epidemic spreads more slowly, and in the limiting case $\sigma \rightarrow 0$, the infection fails to spread at all.

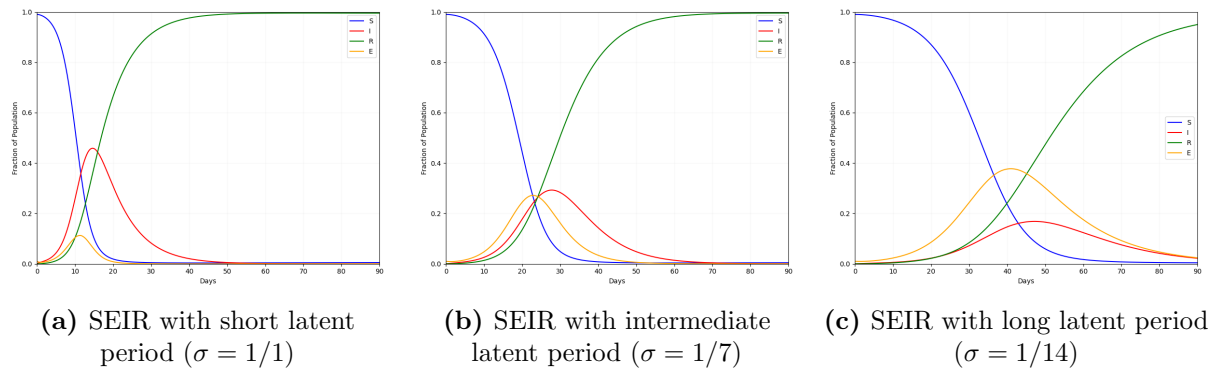


Figure 11: Epidemic dynamics under different latent periods. Longer latent periods (smaller σ) slow down the epidemic and reduce peak prevalence.

4.3.2 Seasonal Forcing

Another important extension is the inclusion of seasonal forcing in the transmission rate β , which now becomes time-dependent, $\beta(t)$, as described in Section 12. Seasonal forcing reflects the fact that transmission intensity fluctuates periodically, due to school terms, climate, or human behavior.

Here, we consider a frequent seasonality period over one year, divided into monthly cycles, to obtain interpretable plots. We set the baseline transmission rate to $\beta_0 = 0.2$, and vary the amplitude of seasonality β_1 , from values close to 0 (weak forcing, baseline-like dynamics) to values close to 1 (strong forcing). The initial condition assumes a mostly susceptible population with a small fraction exposed.

As shown in Figure 12, when β_1 is small (12a), the trajectories of S , E , I , and R remain similar to the basic SEIR dynamics. As β_1 increases (12b and 12c), oscillations become more evident, creating subsequent outbreaks overtime.

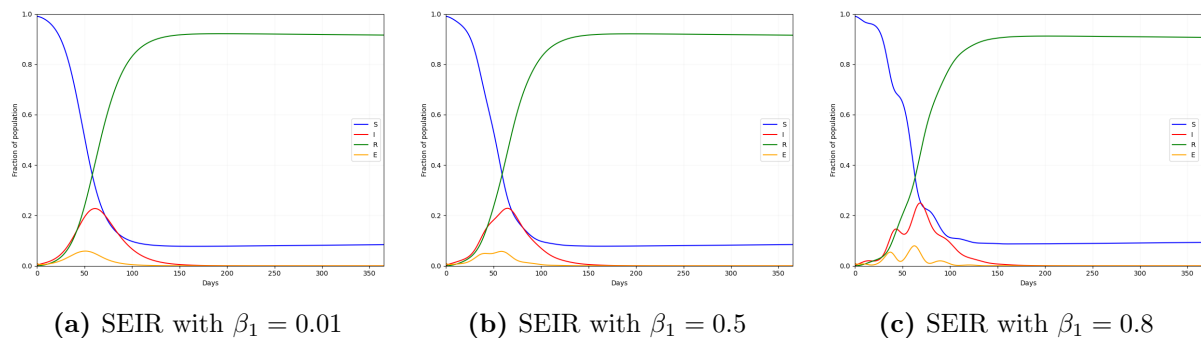


Figure 12: Increasing β_1 strengthens the amplitude of seasonal forcing, producing more evident oscillations in the S , E , I , and R trajectories.

Overall, the dynamics of $\beta(t)$ are controlled by two parameters: the amplitude β_1 , which regulates the strength of oscillations, and the frequency ω , which sets the periodicity of the forcing. Figure 13 illustrates how different combinations of β_1 and ω shape the seasonal variation in $\beta(t)$.

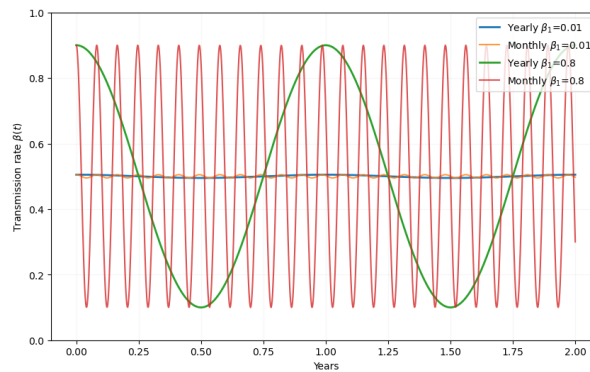


Figure 13: Seasonal forcing of the transmission rate $\beta(t)$ under different values of amplitude β_1 and frequency ω .

5 Conclusions

In conclusion, building on a simple SIR model, we progressively modeled more complex epidemic dynamics, while maintaining interpretability. Our analysis showed that: first, the basic reproductive ratio R_0 determines the threshold condition for an epidemic to develop, but outbreaks can be controlled through vaccination strategies. Next, including demography allows for the existence of an endemic equilibrium, where the disease persists at low prevalence, though infection-induced mortality can alter the dynamics. Finally, the basic compartments and parameters can be modified to account for pathogen characteristics, as well as environmental and social behavior factors, as in the SEIR and seasonality approaches.

Overall, deterministic, continuous-time SIR-based models provide meaningful insights into thresholds, equilibria, and stability, while balancing model complexity and accuracy. However, they may fail to capture the role of chance in real-world epidemics. Random events, contact network structures, and heterogeneity in populations all play major roles in shaping epidemic outcomes. This is why epidemiological analysis must be complemented by stochastic and network-based approaches, which are essential for building realistic models and effective preparedness strategies.

References

- [1] European Centre for Disease Prevention and Control. (2024). *Survey report on national seasonal influenza vaccination recommendations and coverage rates in EU/EEA countries: Data from the 2024 ECDC influenza survey, 2021–22 to 2023–24 influenza seasons*. Publications Office. <https://data.europa.eu/doi/10.2900/4063144>
- [2] Global Preparedness Monitoring Board. (2024). *The changing face of pandemic risk: 2024 GPMB pandemic risk report*. World Health Organization. Licence: CC BY-NC-SA 3.0 IGO.
- [3] Harko, T., Lobo, F. S. N., & Mak, M. K. (2014). Exact analytical solutions of the Susceptible-Infected-Recovered (SIR) epidemic model and of the SIR model with equal death and birth rates. *Applied Mathematics and Computation*, 236, 184–194. <https://doi.org/10.1016/j.amc.2014.03.030>
- [4] Keeling, M. J., & Rohani, P. (2008). *Modeling Infectious Diseases in Humans and Animals*. Princeton University Press. <https://doi.org/10.2307/j.ctvc4gk0>
- [5] Kermack, W. O., & McKendrick, A. G. (1927). A contribution to the mathematical theory of epidemics. *Proceedings of the Royal Society of London Series A Containing Papers of*

a Mathematical and Physical Character, 115(772), 700–721. <https://doi.org/10.1098/rspa.1927.0118>

- [6] Strogatz, S. H. (2014). *Nonlinear dynamics and chaos: With applications to physics, biology, chemistry, and engineering*. Hachette UK.
- [7] Trombetta, C. M., Kistner, O., Montomoli, E., Viviani, S., & Marchi, S. (2022). Influenza viruses and vaccines: The role of vaccine effectiveness studies for evaluation of the benefits of influenza vaccines. *Vaccines*, 10(5), 714. <https://doi.org/10.3390/vaccines10050714>